Chapter 2 — Electronic Devices and Circuits

Syllabus coverage (Unit 2, CEN 05/2025 Annexure-VII C): PN Junction diodes, thyristor; Diode and triode circuits; Junction Transistors; Amplifiers; Oscillator; Multivibrator, counters; Rectifiers; Inverter and UPS.

📊 Weightage & Trend Box

Parameter	Detail
Observed weight (CBT-2, 22-04-2025 shift)	**8 out of 50 technical questions ≈ 16%** — the single highest-weight unit
Expected marks contribution	14–18 out of 100 technical marks
Difficulty band	Easy–Moderate; mostly formula-recall, statement-verification and one-step numericals

Topics actually asked in the 22-04-2025 CBT-2 paper (recreated in the PYQ Corner of this chapter):

1. Rectifier comparison — statement question involving **TUF** (Transformer Utilisation Factor)
2. **BJT current relation** IE = IC + IB and α–β conversions
3. **Half-wave rectifier fed through a transformer** — turns-ratio / output-voltage numerical
4. **Bridge vs centre-tapped** full-wave rectifier — comparative statement question
5. **Colpitts oscillator** — pick-the-incorrect-statement
6. **n-type semiconductor** — majority carriers
7. **Ideal diode circuit** with input V(t) = $\sin^2 t + \cos^2 t$ (a disguised 1 V DC source!)
8. **Oscillator / Barkhausen criterion** — pick-the-false-statement

Preparation strategy: Master the rectifier comparison table (ripple factor, efficiency, PIV, TUF, form factor) until you can reproduce it from memory; memorise every oscillator frequency formula and its identification trick; drill α/β/γ conversions; and never forget the trigonometric identity $\sin^2\theta + \cos^2\theta = 1$ — RRB literally tested it.

2.1 Semiconductor Physics — The Foundation

2.1.1 Energy bands and classification of materials

Electrons in an isolated atom occupy discrete energy levels. When atoms come close in a crystal, these levels split into closely spaced **energy bands**. The two bands that decide electrical behaviour are:

- **Valence band (VB):** highest band filled with electrons at 0 K.
- **Conduction band (CB):** the next higher band; electrons here are free to conduct.
- **Forbidden energy gap (Eg):** separation between the top of VB and bottom of CB.

Material	Energy gap Eg	Behaviour
Conductor (Cu, Al)	Bands overlap, Eg ≈ 0	Abundant free electrons
Semiconductor (Ge, Si, GaAs)	Ge ≈ 0.72 eV, Si ≈ 1.1 eV, GaAs ≈ 1.43 eV	Conduction possible with thermal/optical energy
Insulator (diamond, mica)	Eg > 3 eV (diamond ≈ 6 eV)	Practically no conduction

Key facts repeatedly asked:

- Silicon (Z = 14) and Germanium (Z = 32) are **tetravalent** (4 valence electrons), Group-14 (IV-A) elements forming a diamond-cubic covalent lattice.
- Resistivity order: conductor (10^{-8} Ω·m) < semiconductor (10^{-5}–10^{6} Ω·m) < insulator (10^{11}–10^{19} Ω·m).
- Semiconductors have a **negative temperature coefficient of resistance** — resistance *decreases* as temperature rises because carrier concentration increases exponentially.

2.1.2 Intrinsic semiconductors

A chemically pure semiconductor is **intrinsic**. At 0 K it behaves as a perfect insulator. At room temperature, thermal energy breaks some covalent bonds, generating **electron–hole pairs (EHP)**:

- Free electron → jumps to conduction band.
- **Hole** → the vacancy left in the valence band; behaves as a mobile positive charge carrier.

In an intrinsic semiconductor:

$n = p = n_i$

where n_i is the intrinsic carrier concentration (Si: $n_i \approx 1.5 \times 10^{16}$ /m^3; Ge: $n_i \approx 2.5 \times 10^{19}$ /m^3 at 300 K).

Intrinsic concentration rises steeply with temperature:

$n_i^2 = A_0\, T^3\, e^{\wedge}(-Eg/kT)$

Total current = electron current + hole current. Conductivity of an intrinsic sample:

$\sigma_i = n_i\, e\, (\mu_n + \mu_p)$

where μ_n, μ_p are electron and hole mobilities (for Si: $\mu_n \approx 0.135$ m^2/V·s, $\mu_p \approx 0.048$ m^2/V·s — electrons are roughly 2–3 times more mobile than holes).

2.1.3 Extrinsic semiconductors — doping

Adding a controlled impurity (≈ 1 part in 10^6) enormously increases conductivity. Two types:

n-type (donor doping)

- Dopant: **pentavalent** atom — P, As, Sb, Bi (Group 15/V-A).
- Four electrons bond covalently; the fifth is loosely bound (~0.01–0.05 eV below CB) and is easily donated → **donor level just below the conduction band**.
- **Majority carriers: electrons. Minority carriers: holes.** ← *asked verbatim in the 2025 paper*
- n ≈ N_D (donor concentration); the crystal remains **electrically neutral** overall.

p-type (acceptor doping)

- Dopant: **trivalent** atom — B, Al, Ga, In (Group 13/III-A).
- Only three bonds complete; the vacancy accepts an electron → **acceptor level just above the valence band**.
- **Majority carriers: holes. Minority carriers: electrons.**
- p ≈ N_A.

Mass-action law (holds for both intrinsic and extrinsic material at thermal equilibrium):

$n \cdot p = n_i^2$

So doping that raises one carrier type suppresses the other proportionally.

Charge neutrality: p + N_D = n + N_A.

2.1.4 Drift and diffusion currents

Two transport mechanisms exist in semiconductors:

Drift current — carriers move under an applied electric field E:

- Drift velocity: v_d = μE
- Drift current density: **$J_{drift} = (n e \mu_n + p e \mu_p) E = \sigma E$** (Ohm's law in point form)

Diffusion current — carriers move from high to low concentration (no field needed); unique to semiconductors:

- **$J_n(diff) = +e D_n (dn/dx)$ $J_p(diff) = -e D_p (dp/dx)$**

D_n, D_p are diffusion coefficients linked to mobility by the **Einstein relation**:

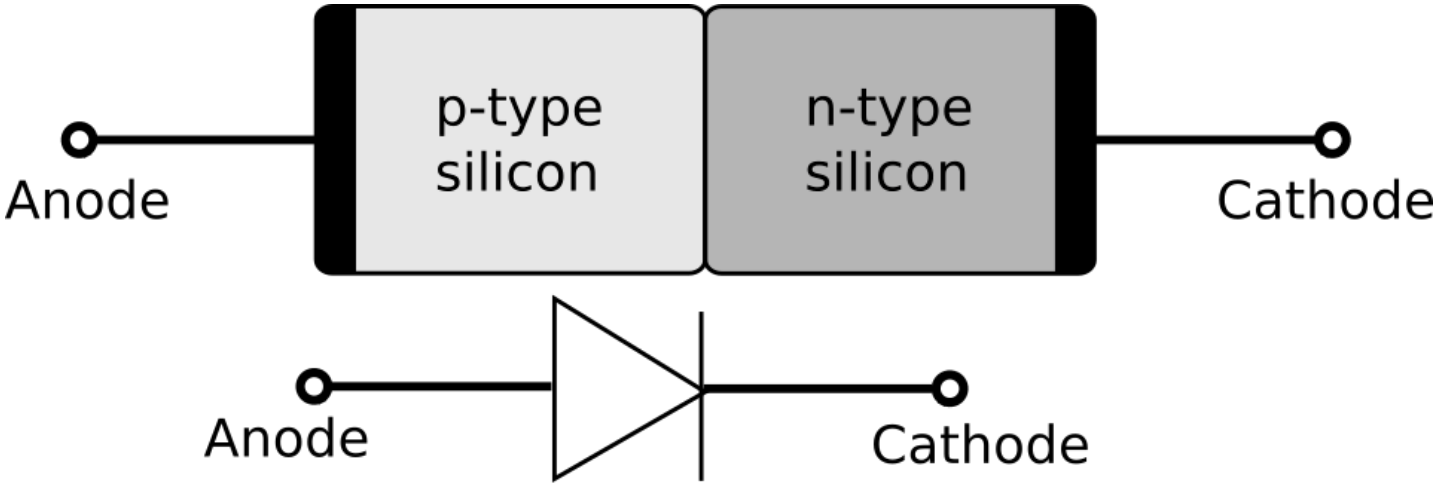
$D/\mu = kT/e = V_T$ (thermal voltage; **$V_T \approx 26$ mV at 300 K**, ≈ T/11600 volts)

Total current density: J = J_drift + J_diffusion for each carrier type.

Hall effect (quick recall): when a current-carrying semiconductor is placed in a transverse magnetic field, a voltage develops perpendicular to both — used to find carrier type and concentration. Hall voltage V_H = BI/(n e t). The polarity of V_H tells whether the sample is n-type or p-type.

2.2 The PN Junction Diode

2.2.1 Junction formation and the depletion region



PN-junction diode: depletion region and circuit symbol.

When p-type and n-type regions are formed in a single crystal:

1. Electrons diffuse from n → p; holes diffuse from p → n (concentration gradients).
2. Near the junction, they recombine, leaving **immobile ions**: negative acceptor ions on the p-side, positive donor ions on the n-side.
3. This carrier-free zone is the **depletion (space-charge) region**, width ≈ 0.1–0.5 μm.
4. The ionic charges create an internal electric field (n → p) and a **barrier potential V_B**:

- **Si: ≈ 0.7 V, Ge: ≈ 0.3 V** (at 25 °C); GaAs ≈ 1.2 V.

- V_B decreases by about **2 mV per °C rise** in temperature.

1. Equilibrium: diffusion current is exactly balanced by drift of minority carriers — net current zero.

Built-in potential (Boylestad/Millman level):

$V_0 = V_T \ln(N_A N_D / n_i^2)$

Depletion width $W \propto \sqrt{(V_0 \pm V)}$ — widens under reverse bias, narrows under forward bias. The region penetrates deeper into the **lightly doped side** ($W \propto 1/N$).

2.2.2 Biasing

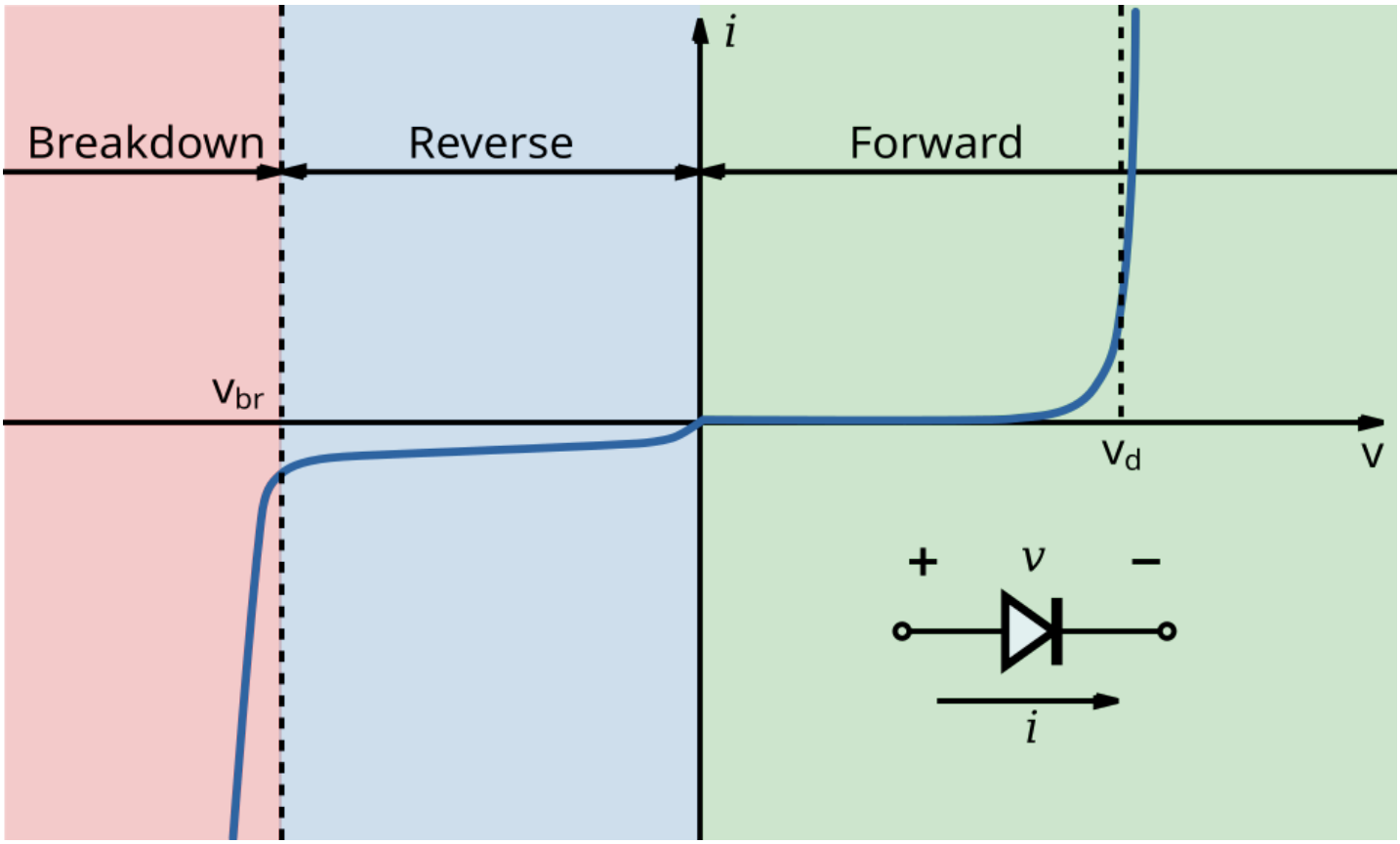
Forward bias (p to +, n to −):

- Applied field opposes barrier; depletion narrows.
- Significant current flows once V > cut-in (knee) voltage **$V_\gamma \approx 0.7$ V (Si) / 0.3 V (Ge)**.
- Current is due to **majority carriers**; grows exponentially.

Reverse bias (p to −, n to +):

- Barrier increases; depletion widens.
- Only a tiny **reverse saturation current I_0 (I_s)** flows, due to **minority carriers**; nearly independent of voltage.
- I_0 **doubles for every 10 °C rise** in temperature (both Si and Ge). I_0 is of order nA for Si, μA for Ge.

2.2.3 Diode equation and V-I characteristics



Diode forward/reverse V–I characteristic.

Shockley diode equation:

$I = I_0 (e^{(V/\eta V_T)} - 1)$

- η (ideality factor) = 1 for Ge, ≈ 2 for Si at low currents (often taken as 1 in problems unless stated).
- $V_T = kT/e \approx 26$ mV at 300 K.

The forward characteristic is exponential after the knee; the reverse characteristic is a flat $-I_0$ line until breakdown.

Diode resistances:

- **Static (DC) resistance:** $R_D = V/I$ at the operating point.
- **Dynamic (AC) resistance:** $r_d = \Delta V/\Delta I = \eta V_T / I$ → the famous **$r_d \approx 26$ mV / I(mA) ohms** (for η = 1).
- Bulk/body resistance of semiconductor material adds in series at high currents.

2.2.4 Diode equivalent models

Model	Description	Use
Ideal diode	Short when forward, open when reverse; $V_\gamma = 0$, $r_f = 0$	Quick circuit analysis, exam "ideal diode" questions
Constant-voltage / practical (2nd approx.)	Battery V_γ (0.7 V Si) in series with ideal diode	Most numericals
Piecewise-linear (3rd approx.)	V_γ + forward resistance r_f in series	Accurate rectifier calculations

2.2.5 Breakdown — Zener vs Avalanche

Under sufficient reverse voltage, the diode conducts heavily — **breakdown**. Not destructive if current is limited.

Feature	Zener breakdown	Avalanche breakdown
Mechanism	High field (~10^7 V/m) directly ruptures covalent bonds — quantum-mechanical tunnelling of electrons	Minority carriers accelerated by field cause impact ionisation → carrier multiplication (chain reaction)
Doping	**Heavily doped**, thin depletion layer	Lightly doped, wide depletion layer
Breakdown voltage	Low: **V_Z < ~5–6 V**	Higher: **V_Z > ~6–8 V**
Temperature coefficient	**Negative** (V_Z falls as T rises)	**Positive** (V_Z rises as T rises)
V-I knee	Sharp	Softer

Around 5–6 V both mechanisms coexist and the temperature coefficient is nearly zero — hence 5.6 V Zeners are the most temperature-stable.

2.2.6 Zener diode as voltage regulator

A Zener operated in reverse breakdown maintains nearly constant voltage V_Z across itself. Standard shunt-regulator circuit: unregulated V_{in} → series resistor R_S → Zener (in reverse) in parallel with load R_L.

Analysis (Zener ON when the open-circuit voltage across it ≥ V_Z):

- Voltage across load: **$V_L = V_Z$**
- Series-resistor current: **$I_S = (V_{in} - V_Z)/R_S$**
- Load current: **$I_L = V_Z/R_L$**
- Zener current: **$I_Z = I_S - I_L$** (must satisfy $I_{Z(min)} \le I_Z \le I_{Z(max)} = P_Z/V_Z$)

Regulation works against both **line variations** (changing V_{in}) and **load variations** (changing R_L). Worst cases: I_Z is maximum when V_{in} is max and I_L is min; I_Z is minimum when V_{in} is min and I_L is max.

2.2.7 Special-purpose diodes

Diode	Bias used	Principle	Applications / key point
LED	Forward	Electroluminescence — radiative recombination of injected carriers	Made from **GaAs (IR), GaAsP (red/yellow), GaP (green)** — never plain Si/Ge (indirect band gap). λ = 1240/Eg(eV) nm. Cut-in ≈ 1.8–2.4 V
Photodiode	**Reverse**	Light generates EHPs in depletion region → photocurrent ∝ illumination	Light detectors, optocouplers; faster than LDR
Solar cell	No external bias	Photovoltaic effect at unbiased junction	Large area junction; open-circuit voltage ≈ 0.5–0.6 V per Si cell
Varactor (varicap)	**Reverse**	Depletion width (hence junction/transition capacitance $C_T \propto 1/\sqrt{V_R}$) varies with reverse voltage	Electronic tuning, VCOs, AFC
Schottky (hot-carrier)	Forward	**Metal–semiconductor** junction; conduction by majority carriers only — no minority-carrier storage	Very fast switching, low forward drop ≈ 0.2–0.3 V; used in SMPS, clamping in TTL (Schottky TTL)
Tunnel (Esaki) diode	Forward	Very heavy doping ($1:10^3$) → extremely thin junction → quantum tunnelling	Exhibits **negative resistance region** between peak ($V_P \approx 65$ mV) and valley ($V_V \approx 350$ mV Ge); microwave oscillators, high-speed switching
PIN diode	Reverse/RF	Intrinsic layer between p and n	RF switch/attenuator

Diode capacitances: transition (depletion) capacitance C_T dominates in **reverse** bias (varactor); diffusion (storage) capacitance C_D dominates in **forward** bias and limits switching speed (reverse recovery time t_{rr}).

2.3 Diode Circuits — Rectifiers

Rectification = conversion of AC into unidirectional (pulsating) DC. Input: $v = V_m \sin \omega t$ from the transformer secondary.

2.3.1 Half-wave rectifier (HWR)

One diode conducts only in the positive half-cycle; output is zero for the negative half.

Derivations (ideal diode, resistive load R_L):

- DC (average) load current: $I_{dc} = \frac{1}{2\pi}\int_0^{\pi} I_m \sin\theta \, d\theta = \mathbf{I_m/\pi}$
- DC output voltage: $\mathbf{V_{dc} = V_m/\pi \approx 0.318\ V_m}$
- RMS load current: $I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} I_m^2 \sin^2\theta \, d\theta} = \mathbf{I_m/2}$
- **Form factor** $= I_{rms}/I_{dc} = (I_m/2)/(I_m/\pi) = \mathbf{\pi/2 = 1.57}$
- **Ripple factor** $\gamma = \sqrt{FF^2 - 1} = \sqrt{1.57^2 - 1} = \mathbf{1.21}$
- **Rectification efficiency** $\eta = P_{dc}/P_{ac} = (I_{dc}^2 R_L)/(I_{rms}^2(r_f + R_L)) \rightarrow$ max $\mathbf{\eta = 4/\pi^2 = 40.6\ \%}$ (ideal diode)
- **PIV = V_m**

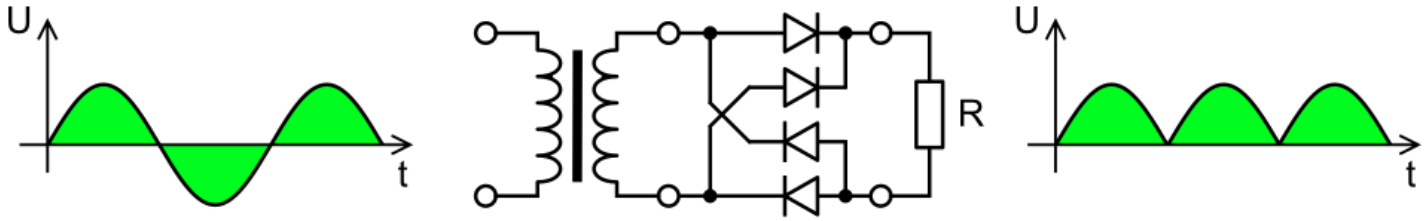
- **TUF = 0.287** (secondary); because DC flows through the secondary, the transformer core suffers **DC saturation** — a major drawback.
- Ripple (lowest/fundamental) frequency = **f** (same as supply, e.g. 50 Hz).

2.3.2 Full-wave centre-tapped rectifier (FWR-CT)

Two diodes + centre-tapped secondary; each half of the secondary drives one diode alternately, so **both half-cycles** appear across the load in the same direction.

- **$I_{dc} = 2I_m/\pi$, $V_{dc} = 2V_m/\pi \approx 0.636\ V_m$** ($V_m$ = peak of *half* secondary)
- **$I_{rms} = I_m/\sqrt{2}$**
- **Form factor = $(I_m/\sqrt{2})/(2I_m/\pi) = \pi/(2\sqrt{2}) = 1.11$**
- **Ripple factor $\gamma = \sqrt{(1.11^2 - 1)} = 0.482 \approx 0.48$**
- **$\eta(max) = 8/\pi^2 = 81.2\ \%$**
- **PIV = $2V_m$** ← the big penalty of the centre-tap topology
- **TUF = 0.693** (average of primary 0.812 and secondary 0.574)
- Ripple frequency = **2f** (100 Hz on 50 Hz mains).

2.3.3 Bridge rectifier



Full-wave bridge rectifier (Graetz bridge).

Four diodes in a bridge; no centre tap required. In each half-cycle two diagonal diodes conduct in series.

- Same waveform maths as FWR-CT: $V_{dc} = 2V_m/\pi$, FF = 1.11, γ = 0.48, $\eta(max)$ = 81.2 %
- **PIV per diode = V_m only** (half that of centre-tap for the same output)
- **TUF = 0.812** — best transformer utilisation
- Two diode drops (≈ 1.4 V for Si) in series each half-cycle — slight disadvantage at low voltages
- Ripple frequency = **2f**.

2.3.4 Master comparison table (memorise completely)

Parameter	Half-wave	Full-wave centre-tap	Bridge
No. of diodes	1	2	4
V_{dc} (ideal)	$V_m/\pi = 0.318\ V_m$	$2V_m/\pi = 0.636\ V_m$	$2V_m/\pi = 0.636\ V_m$
I_{rms}	$I_m/2$	$I_m/\sqrt{2}$	$I_m/\sqrt{2}$
Form factor	**1.57**	**1.11**	**1.11**
Ripple factor	**1.21**	**0.482**	**0.482**
Max efficiency	**40.6 %**	**81.2 %**	**81.2 %**
PIV per diode	V_m	**$2V_m$**	**V_m**
TUF	0.287	0.693	**0.812**
Ripple frequency	f	2f	2f
Centre-tapped transformer?	No	**Yes**	No
Diode drops in path	1	1	2
DC core saturation	Yes	No	No

Bridge vs centre-tap (statement questions love these):

1. Bridge needs no centre-tapped transformer (cheaper, smaller transformer).
2. PIV rating of bridge diodes is half → suits **high-voltage** applications.
3. Bridge has the **highest TUF (0.812)**.
4. Centre-tap wins only where the two extra diode drops matter (very low output voltages) since it has one diode drop per path.

2.3.5 Transformer Utilisation Factor (TUF) — asked in 2025

TUF = P_{dc} delivered to load / AC (VA) rating of the transformer

It measures how effectively transformer VA capacity is converted into DC power. TUF is always < 1. Ranking: **Bridge (0.812) > FW centre-tap (0.693) > Half-wave (0.287)**. Low TUF of HWR means a much bigger transformer is needed for the same DC output.

2.3.6 Filters

The pulsating rectifier output = DC + ripple harmonics. Filters pass DC and block ripple.

Filter	Connection	Behaviour	Ripple factor (FWR)	Best at
Shunt capacitor (C)	C across load	C charges to V_m, feeds load during diode-off time	$\gamma = 1/(4\sqrt{3}\, f\, C\, R_L)$	Light loads (small I_{dc}). Surge current on diodes is high
Series inductor (L)	L in series	L opposes changes in current	$\gamma = R_L/(3\sqrt{2}\,\omega L)$	Heavy loads (large I_{dc})
LC (L-section)	L series + C shunt	Combines both; ripple independent of load	$\gamma = \sqrt{2}/(12\,\omega^2 LC)$	Good regulation; needs bleeder resistor & critical inductance $L_C = R_L/3\omega$
π (CLC)	C–L–C	Highest DC output (≈ V_m), best smoothing	Higher than LC at same L,C values? No — lowest ripple, but **poor regulation**	Low-current, low-ripple loads

Capacitor-filter DC output ≈ $V_m - I_{dc}/(4fC)$ for full-wave. Peak-to-peak ripple $V_{r(pp)} = I_{dc}/(2fC)$ (full-wave), $I_{dc}/(fC)$ (half-wave).

2.3.7 Clippers and clampers

Clipper (limiter): removes part of the waveform above/below a reference level using a diode ± bias battery. Series clipper (diode in series with load) or shunt clipper (diode across load). Output amplitude is *changed*; DC level essentially unchanged. Applications: wave shaping, protection, TV sync separation.

- Positive clipper: removes positive peaks. Biased clipper: clips beyond $V_\gamma + V_{bias}$.
- Two-level (slicer/zener) clipper: clips both peaks — makes an approximate square wave from a sine.

Clamper (DC restorer): shifts the entire waveform up or down **without changing its shape or peak-to-peak value**, using a capacitor + diode (+ optional bias). $RC \gg T$ (typically $\geq 10T$).

- Positive clamper: pushes waveform up (diode conducts on negative peak, charging C to V_m).
- Negative clamper: pushes it down.
- Total swing after clamping = same $2V_m$; new peak = $2V_m$ (voltage doubler principle builds on this).

Rule of thumb: clipper *cuts*, clamper *shifts*.

2.4 Junction Transistor (BJT)

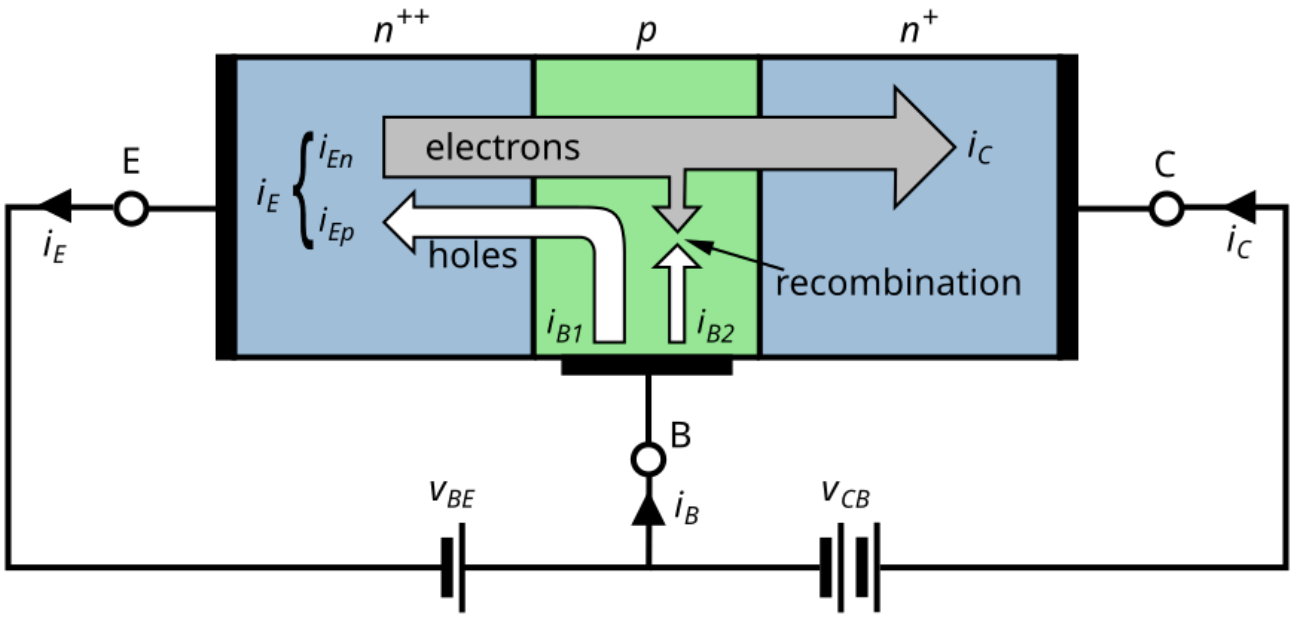
2.4.1 Construction

A Bipolar Junction Transistor = two PN junctions back-to-back sharing a common middle region: **NPN** or **PNP**.

Region	Doping	Width	Role
Emitter (E)	Heaviest	Moderate	Injects (emits) majority carriers
Base (B)	Lightest	**Thinnest** (~μm)	Passes carriers through; small recombination
Collector (C)	Moderate	**Largest area**	Collects carriers; large area to dissipate heat

"Bipolar" = both electrons **and** holes take part in conduction. NPN is preferred: electron mobility > hole mobility → faster, higher gain.

2.4.2 Operation (active region, NPN)



NPN BJT carrier flow in the active region.

Bias for amplifier action: **emitter–base junction forward biased, collector–base junction reverse biased.**

1. Forward-biased EB junction injects electrons from emitter into the base.
2. Base is thin and lightly doped → only ~1–5 % recombine (forming I_B).
3. ~95–99 % diffuse across and are swept into the collector by the reverse-biased CB field → I_C.

Fundamental current relation (KCL on the device):

I_E = I_C + I_B ← asked directly in 2025

Including leakage: I_C = αI_E + I_CBO, and in CE form I_C = βI_B + I_CEO where **I_CEO = (1+β) I_CBO**.

2.4.3 Current gains α, β, γ

- **α (CB current gain)** = I_C/I_E (0.95–0.998, always < 1)
- **β = h_FE (CE current gain)** = I_C/I_B (20–500)
- **γ (CC current gain)** = I_E/I_B = **1 + β**

Conversions (drill these):

β = α/(1−α) α = β/(1+β) γ = 1 + β = 1/(1−α)

2.4.4 Configurations — comparison table

Property	Common Base (CB)	Common Emitter (CE)	Common Collector (CC)
Input between	E–B	B–E	B–C
Current gain	$\alpha < 1$	**β (high)**	$\gamma = 1+\beta$ (highest)
Voltage gain	High	**High**	< 1 (≈ 0.95–0.99)
Power gain	Moderate	**Highest**	Lowest
Input impedance	Very low (20–100 Ω)	Moderate (~1 kΩ)	**Very high (~100s of kΩ)**
Output impedance	Very high (~MΩ)	Moderate (~50 kΩ)	**Very low (~50 Ω)**
Phase shift (in–out)	0°	**180°**	0°
Applications	HF amplifier, impedance step-up	**General amplifier** (most used)	Buffer / emitter follower, impedance matching

2.4.5 CE output characteristics and regions of operation

Plot of I_C vs V_{CE} for steps of I_B shows three regions:

Region	EB junction	CB junction	Behaviour	Use
Cut-off	Reverse	Reverse	$I_C \approx 0$ (only leakage)	Switch OFF
Active	Forward	Reverse	$I_C = \beta I_B$, nearly flat curves	**Amplifier**
Saturation	Forward	Forward	$V_{CE} \approx 0.2$ V (Si), I_C limited by external circuit	Switch ON

Early effect (base-width modulation): increasing V_{CE} slightly narrows the base → curves have small upward slope in active region.

2.4.6 DC load line and Q-point

For a CE stage with collector supply V_{CC} and collector resistor R_C (plus R_E if present), KVL gives:

$V_{CE} = V_{CC} - I_C (R_C + R_E)$

This straight line on the output characteristics is the **DC load line**:

- V_{CE}-axis intercept (cut-off point): $V_{CE} = V_{CC}$ ($I_C = 0$)
- I_C-axis intercept (saturation point): $I_{C(sat)} = V_{CC}/(R_C + R_E)$

The **quiescent (Q) point** is the intersection of the load line with the $I_B = I_{BQ}$ characteristic. For maximum symmetrical swing in class-A: place Q at the **middle** of the load line ($V_{CEQ} \approx V_{CC}/2$).

AC load line has slope $-1/(R_C \parallel R_L)$ and passes through Q; it is steeper than the DC load line.

2.4.7 Biasing methods and stability factor

Bias circuits must hold Q fixed against temperature (I_{CBO} doubles/10 °C, V_{BE} −2 mV/°C, β spread).

Stability factor $S = \partial I_C / \partial I_{CBO}$ (smaller is better; ideal = 1).

Method	Circuit essence	S	Comment
Fixed bias (base bias)	Single R_B from V_CC to base	**S = 1 + β** (worst)	Simplest; thermally unstable; risk of thermal runaway
Collector-to-base bias	R_B from collector to base	S = (1+β)/(1 + β·R_C/(R_C+R_B))	Negative feedback improves stability; reduces gain
Emitter bias (fixed + R_E)	R_B + emitter resistor R_E	Between fixed and voltage-divider	R_E gives current-series feedback
Voltage-divider (self) bias	R_1–R_2 divider + R_E	**S = (1+β)·(1 + R_B/R_E)/(1 + β + R_B/R_E) ≈ 1 + R_B/R_E** (best practical)	Industry standard; Q nearly independent of β

Voltage-divider analysis: V_B = V_CC·R_2/(R_1+R_2) (if divider stiff, i.e. I_divider ≫ I_B); V_E = V_B − 0.7; I_C ≈ I_E = V_E/R_E; V_CE = V_CC − I_C(R_C + R_E).

Thermal runaway: rise in T → I_CBO↑ → I_C↑ → junction power ↑ → T↑ … cumulative destruction. Prevented by R_E (self-bias), heat sinks, and ensuring V_CE < V_CC/2 condition for thermal stability.

2.4.8 BJT as a switch

- **OFF:** base open/reverse biased → cut-off → V_out ≈ V_CC.
- **ON:** enough base drive to saturate (I_B > I_C(sat)/β) → V_CE(sat) ≈ 0.2 V.

Used in logic gates, relay/LED drivers, choppers. Switching speed is limited by storage time (charge stored in base during saturation).

2.5 Field-Effect Transistors — JFET and MOSFET (basics)

FETs are **unipolar** (conduction by one carrier type only), **voltage-controlled** devices with very high input impedance.

2.5.1 JFET

- Structure: n-channel (or p-channel) bar with two p^+ (or n^+) gate regions; terminals **Source, Gate, Drain**.
- Gate–channel junction always **reverse biased**; input impedance ~10^8 Ω.
- Increasing |V_GS| widens depletion, narrows channel; at **pinch-off voltage V_P**, channel closes and I_D saturates.
- Transfer (Shockley) equation: **$I_D = I_{DSS}(1 - V_{GS}/V_P)^2$** — square-law device (low intermodulation distortion).
- Transconductance: **$g_m = \partial I_D/\partial V_{GS} = g_{m0}(1 - V_{GS}/V_P)$**, where $g_{m0} = 2I_{DSS}/|V_P|$.
- Amplification factor: **μ = g_m × r_d**.
- JFET is a **depletion-mode-only** device; normally ON at V_GS = 0.

2.5.2 MOSFET

Metal-Oxide-Semiconductor FET: gate insulated from channel by SiO_2 → input impedance 10^{10}–10^{15} Ω; handle with anti-static care (gate oxide punctures easily).

Type	State at V_GS = 0	Control
Depletion MOSFET (D-MOSFET)	ON (channel built in)	Can be operated in both depletion and enhancement modes
Enhancement MOSFET (E-MOSFET)	**OFF** (no channel)	Conducts only when V_GS > threshold V_T; $I_D = k(V_{GS} - V_T)^2$

E-MOSFETs (NMOS/PMOS) are the basis of **CMOS** digital ICs — negligible static power. Power MOSFETs (VDMOS) dominate SMPS and motor drives.

BJT vs FET quick contrast: BJT — bipolar, current-controlled, low input Z, higher gain, cheaper; FET — unipolar, voltage-controlled, very high input Z, better thermal stability (negative temperature coefficient of I_D prevents thermal runaway), lower noise.

2.6 Transistor Amplifiers

2.6.1 Small-signal CE amplifier

The workhorse voltage amplifier: voltage-divider-biased CE stage with coupling capacitors (block DC) and emitter bypass capacitor (restores AC gain).

- AC emitter resistance: **$r_e' = 26\ mV / I_{E(mA)}$**
- Voltage gain (bypassed R_E): **$A_v \approx -(R_C \parallel R_L)/r_e'$** (minus = 180° phase inversion)
- Input impedance: $Z_{in} = R_1 \parallel R_2 \parallel \beta\, r_e'$
- Removing the bypass capacitor: $A_v \approx -R_C/(r_e' + R_E)$ — lower but stabler gain (swamping).

2.6.2 h-parameters (hybrid model) — basics

For any two-port: $v_1 = h_i\, i_1 + h_r\, v_2$; $i_2 = h_f\, i_1 + h_o\, v_2$.

Symbol (CE)	Name	Typical value	Unit
h_ie	Input impedance (output shorted)	1–2 kΩ	Ω
h_re	Reverse voltage ratio	2.5×10^{-4}	—
h_fe (= β_ac)	Forward current gain	50–300	—
h_oe	Output admittance (input open)	25 µS	siemens (℧)

Note the mixed units → "hybrid". Approximate CE results: $A_i = -h_{fe}/(1 + h_{oe} R_L) \approx -h_{fe}$; $Z_{in} \approx h_{ie}$; $A_v = -h_{fe} R_L/h_{ie}$.

2.6.3 Multistage & RC-coupled amplifier frequency response

Cascading raises gain: overall $A = A_1 \times A_2 \times \ldots$; in dB, gains add. Coupling methods: RC (cheapest, audio), transformer (impedance matching, power), direct (DC amplifiers, drift problem).

RC-coupled amplifier response (the classic curve):

- **Low-frequency roll-off:** coupling & bypass capacitors' reactance rises → gain falls.
- **Mid-band:** capacitors ≈ shorts, strays ≈ open → flat maximum gain.
- **High-frequency roll-off:** shunt/stray/junction capacitances (and Miller capacitance $C_M = C_{bc}(1+|A_v|)$) short the signal.
- **Cut-off frequencies f_1 (lower), f_2 (upper):** gain = $A_{mid}/\sqrt{2}$ = **−3 dB points** (half-power points).
- **Bandwidth $BW = f_2 - f_1$; Gain × Bandwidth = constant** (figure of merit f_T).

2.6.4 Amplifier classes (power amplifiers)

Class	Conduction angle	Q-point location	Max theoretical efficiency	Distortion	Use
A	360°	Centre of load line	**25 %** (series-fed), **50 %** (transformer-coupled)	Lowest	Small-signal, hi-fi
B	180°	At cut-off	**78.5 % (π/4)**	Crossover distortion in push-pull	Push-pull output stages
AB	180°–360°	Slightly above cut-off	50–78.5 %	Removes crossover distortion	Audio power amps
C	< 180°	Below cut-off	> 90 % (up to ~99 %)	Very high — needs tuned tank	RF tuned amplifiers, oscillators

Push-pull class-B: two transistors handle alternate half-cycles; even harmonics cancel. Complementary-symmetry version (NPN + PNP) eliminates transformers. Crossover distortion (dead zone ±0.7 V) removed by class-AB pre-bias (diode biasing).

2.6.5 Distortion & noise (one-liners)

- Amplitude (harmonic) distortion — nonlinearity of characteristics; measured by THD.
- Frequency distortion — unequal gain at different frequencies.
- Phase distortion — unequal phase shift vs frequency.
- Noise figure NF = (S/N)_in / (S/N)_out in dB; ideal amplifier NF = 0 dB.

2.7 Feedback in Amplifiers

Feedback returns a fraction β_f (feedback factor) of the output to the input.

Closed-loop gain: A_f = A / (1 + Aβ_f) (negative feedback; loop gain = Aβ_f)

If |1 + Aβ_f| > 1 → **negative (degenerative)**; if < 1 → positive (regenerative); if 1 + Aβ_f = 0 → oscillation.

Effects of negative feedback (all by the factor D = 1 + Aβ_f):

Quantity	Effect
Gain	Decreases: A_f = A/D
Gain stability	Improves: dA_f/A_f = (1/D)(dA/A)
Bandwidth	**Increases**: BW_f = D × BW (gain × BW constant)
Distortion & noise (in loop)	Decreases by D
Input impedance	↑ for **series** mixing; ↓ for shunt mixing
Output impedance	↓ for **voltage** (shunt) sampling; ↑ for current (series) sampling

Four topologies:

Topology (sampling–mixing)	Alt name	Z_in	Z_out	Stabilised gain	Example
Voltage-series	Series-shunt	↑	↓	Voltage gain	Emitter follower, non-inverting op-amp
Voltage-shunt	Shunt-shunt	↓	↓	Transresistance	Collector-to-base bias, inverting op-amp
Current-series	Series-series	↑	↑	Transconductance	Unbypassed R_E stage
Current-shunt	Shunt-series	↓	↑	Current gain	Two-stage feedback pair

When $A\beta_f \gg 1$: **$A_f \approx 1/\beta_f$** — gain set only by (passive, stable) feedback network.

2.8 Oscillators

An oscillator converts DC into AC of desired frequency **without any external input signal** — it is an amplifier with **positive feedback** satisfying:

2.8.1 Barkhausen criterion — asked in 2025

For sustained oscillations:

1. **Loop gain magnitude $|A\beta_f| = 1$** (in practice slightly > 1 to start; amplitude then self-limits), and
2. **Total loop phase shift = 0° or 360°** (i.e. $2n\pi$).

The starting "input" is internal noise/switching transient — no external signal is required. If $|A\beta| < 1$ oscillations die out; if held $\gg 1$ the output distorts/clips.

2.8.2 RC oscillators (audio frequency)

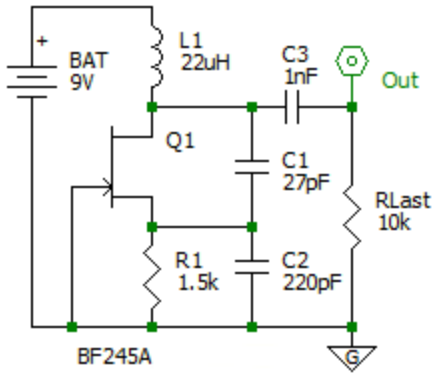
RC phase-shift oscillator

- CE amplifier (180°) + three-section RC ladder giving another 180° (60° per section).
- **$f = 1/(2\pi RC\sqrt{6})$** (FET/ideal version); with BJT loading: $f = 1/[2\pi RC\sqrt{(6 + 4R_C/R)}]$.
- Requires amplifier gain **$|A| \geq 29$** ($\beta_f = 1/29$).

Wien bridge oscillator

- Series RC + parallel RC (lead–lag) network in a bridge; two-stage/non-inverting amplifier (0° amplifier phase, 0° network phase at f_0).
- **$f = 1/(2\pi RC)$** (equal R, C).
- Network attenuation = 1/3 → requires **gain ≥ 3** (so in op-amp form $R_f = 2R_1$).
- Amplitude stabilised by a thermistor/sensistor or lamp. Excellent low-distortion sine source; standard in audio signal generators.

2.8.3 LC oscillators (radio frequency)



Colpitts oscillator (capacitive tap — an LC oscillator).

General form: tank circuit of reactances X_1, X_2 (across the two amplifier input/output nodes) and X_3 (feedback). Oscillation needs X_1, X_2 same type and X_3 opposite type.

Hartley oscillator — **tapped inductor** (two inductors L_1, L_2, may be mutually coupled M) + one capacitor:

$f = 1/(2\pi\sqrt{(L_{eq} C)})$, $L_{eq} = L_1 + L_2 + 2M$; feedback fraction $\beta_f = L_2/L_1$ (gain needed ≈ L_1/L_2).

Colpitts oscillator — **tapped capacitor** (C_1, C_2 in series) + one inductor:

$f = 1/(2\pi\sqrt{(L\, C_{eq})})$, $C_{eq} = C_1C_2/(C_1 + C_2)$; feedback fraction = C_1/C_2 (voltage divider across capacitors); gain condition ≈ C_2/C_1 (using amplifier input side C_2). Better frequency stability than Hartley at high f; capacitors are cheaper/purer than tapped coils.

Clapp oscillator — Colpitts + small capacitor **C_3 in series with L**; $C_3 \ll C_1$, C_2 dominates the frequency:

$f = (1/2\pi)\sqrt{[(1/L)(1/C_1 + 1/C_2 + 1/C_3)]} \approx 1/(2\pi\sqrt{(LC_3)})$

Frequency almost independent of transistor junction capacitances → improved stability.

Identification trick: Colpitts → **C**apacitive divider; **H**artley → tapped **H**enries (inductor); Clapp → Colpitts + extra series C.

2.8.4 Crystal oscillator

Quartz crystal = electromechanical resonator (piezoelectric effect). Equivalent circuit: series R-L-C (motional arm) in parallel with mount capacitance C_M.

- **Series resonance $f_s = 1/(2\pi\sqrt{(LC)})$**; parallel (antiresonance) f_p slightly above f_s; crystal is **inductive between f_s and f_p** — the region used in oscillator circuits (e.g. Pierce = crystal Colpitts).
- Extraordinary **Q (10^4–10^6)** → frequency stability ~1 ppm or better.
- Used wherever accuracy matters: clocks, microprocessors, transmitters, watches.

Frequency-stability ranking: Crystal ≫ Clapp > Colpitts ≈ Hartley > RC types.

2.8.5 Negative-resistance & relaxation oscillators

- Tunnel-diode and UJT oscillators exploit negative-resistance/relaxation action (UJT covered in §2.10.6).
- Relaxation oscillators produce **non-sinusoidal** (sawtooth/square) outputs via RC charge–discharge.

2.9 Multivibrators and Counters

Multivibrators are two-stage regenerative (positive-feedback) switching circuits producing square/rectangular waves. Each stage drives the other; the transistors act as switches, alternately cut-off and saturated.

2.9.1 The three types

Type	Stable states	Coupling (classic BJT circuit)	Output	Also called	Applications
Astable	**0** (free-running)	Both stages **capacitor (AC)** coupled	Continuous square wave	Free-running MV	Clock generation, flashers
Monostable	**1**	One capacitive + one resistive (DC) coupling	Single pulse of fixed width per trigger	One-shot	Pulse-width generation, timing/delay
Bistable	**2**	Both stages **resistive (DC)** coupled	Toggles state on each trigger, holds indefinitely	Flip-flop, Eccles–Jordan, binary	Memory, **counters**, frequency division

2.9.2 Timing formulas (BJT collector-coupled circuits)

- Half-period of astable per side: T_1 = **$0.693\ R_1C_1$** (time for the capacitor to bring the off transistor's base up through V_BE); total **$T = 0.693(R_1C_1 + R_2C_2)$**; symmetric case **$T = 1.386\ RC$**, f = 1/T ≈ **$0.72/RC$**.
- Monostable pulse width: **$T = 0.693\ RC$**.
- (IC 555 versions: astable T = 0.693(R_A + 2R_B)C; monostable T = 1.1 RC — useful cross-checks.)

2.9.3 Schmitt trigger

An emitter-coupled bistable with **hysteresis**: two different threshold voltages, Upper Trip Point (UTP) and Lower Trip Point (LTP). Converts any slowly varying/noisy waveform into a clean square wave — a "squaring circuit"/level detector. Hysteresis width = UTP − LTP gives noise immunity.

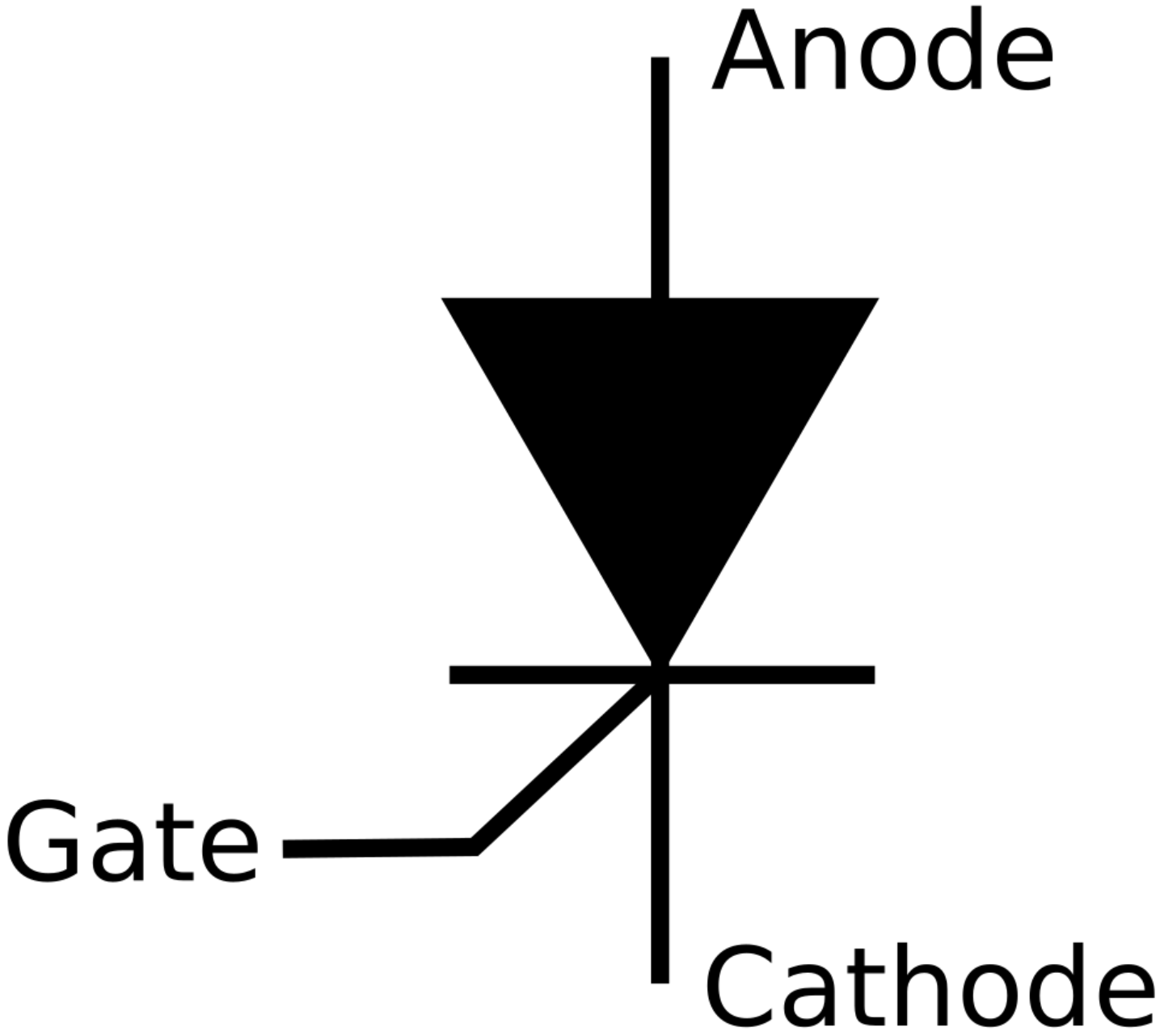
2.9.4 Counters (as required by the syllabus keyword)

A **counter** is a chain of bistable multivibrators (flip-flops). Each flip-flop **divides frequency by 2** (one output pulse for every two input triggers).

- **n flip-flops → modulus up to 2^n** (counts 0 to $2^n - 1$). For MOD-m: n = $\lceil \log_2 m \rceil$ flip-flops; unwanted states skipped by reset logic (e.g. decade counter MOD-10 uses 4 FFs, resets at 1010).
- **Asynchronous (ripple) counter:** clock ripples through FFs; simple but slow (delays add: t_total = n·t_pd).
- **Synchronous counter:** common clock to all FFs; faster, glitch-free.
- **Ring counter:** n FFs → MOD-n; **Johnson (twisted-ring):** n FFs → MOD-2n.
- Applications: digital clocks, frequency division/measurement, event counting. (Full digital treatment appears in the Digital Electronics chapter; here remember the multivibrator–counter link: **bistable = 1-bit memory = divide-by-2**.)

2.10 Thyristor Family

2.10.1 SCR — construction



SCR / thyristor symbol.

The Silicon Controlled Rectifier is a **four-layer (P-N-P-N), three-junction (J_1, J_2, J_3), three-terminal** device: **Anode (A), Cathode (K), Gate (G — attached to the inner p-layer near the cathode)**. It is a unidirectional, latching power switch (ratings to several kA / kV).

2.10.2 V-I characteristic

- **Reverse blocking (A negative):** J_1, J_3 reverse biased → tiny leakage until reverse breakdown V_BR. SCR must never be gated here.
- **Forward blocking (A positive, no gate):** J_2 reverse biased → only leakage flows (OFF state) until the **forward break-over voltage V_BO**.
- **Forward conduction (ON):** once triggered (gate pulse or V > V_BO), J_2 collapses; device latches like a closed switch, drop ≈ 1–2 V. **Increasing gate current lowers the break-over voltage.**

- **Latching current I_L:** minimum anode current to latch ON just after triggering (gate may then be removed).
- **Holding current I_H:** minimum anode current below which the SCR turns OFF. **I_L > I_H** (typically I_L ≈ 2–3 × I_H).

2.10.3 Two-transistor analogy

Split the PNPN structure into a PNP (Q_1) and an NPN (Q_2) transistor: collector of each feeds the base of the other. Anode current:

$I_A = (\alpha_1 + \alpha_2) I_A + I_{leak} \rightarrow I_A = I_{leak} / [1 - (\alpha_1 + \alpha_2)]$

When gate/leakage current raises $\alpha_1 + \alpha_2 \rightarrow 1$, $I_A \rightarrow \infty$ (limited only by external circuit) — **regenerative latching**. This explains: (i) why the gate loses control after turn-on, and (ii) dv/dt false triggering (junction capacitance current mimics gate current).

2.10.4 Triggering (turn-on) and commutation (turn-off)

Turn-on methods:

1. **Gate triggering** (normal method — small gate pulse while forward biased)
2. Forward-voltage triggering (exceeding V_BO — avoided in practice)
3. **dv/dt triggering** (fast anode voltage rise — unwanted; suppressed by RC **snubber**)
4. Temperature triggering (leakage rise — unwanted)
5. Light triggering (**LASCR** — used in HVDC valves)

Turn-off (commutation) — anode current must fall below I_H and reverse bias/zero bias held for the turn-off time t_q:

- **Natural (line) commutation:** AC supply reverses every half-cycle — free of cost; used in phase-controlled rectifiers, AC regulators.
- **Forced commutation:** in DC circuits an auxiliary L-C circuit forces current to zero (classes A–F) — used in choppers and (older) inverters.

Gate protection & firing: series resistor limits gate current, R-C across gate for noise, pulse-transformer/optocoupler isolation; R-firing, RC-firing and UJT-firing circuits give 0–90°, 0–180°, and precise wide-range control respectively.

2.10.5 TRIAC and DIAC

TRIAC = two SCRs in inverse parallel integrated in one 5-layer chip; terminals **MT_1, MT_2, Gate**. Conducts in **both directions**; can be triggered by positive or negative gate pulses (four triggering modes; quadrants I+ and III− are most sensitive). Used in AC phase control: fan regulators, lamp dimmers, temperature control. Drawback vs two SCRs: lower dv/dt rating, less sensitive in some quadrants.

DIAC = gateless bidirectional trigger diode (3-layer/5-layer symmetric). Blocks until break-over (≈ 30–35 V) in either polarity, then exhibits negative resistance. Almost always used in the gate circuit of a TRIAC (dimmers) to deliver sharp symmetric trigger pulses.

Other members (one-liners): GTO — gate turn-off thyristor (negative gate pulse turns it OFF; no forced-commutation circuit needed); SCS — silicon controlled switch (two gates); PUT — programmable UJT; MCT/IGBT covered in Power Electronics.

2.10.6 UJT and the relaxation oscillator

UJT (Uni-Junction Transistor): lightly doped n-type silicon bar with base contacts B_1, B_2 and a single p-emitter part-way along — one PN junction, hence the name; also called *double-base diode*.

- **Intrinsic stand-off ratio: $\eta = R_{B1}/(R_{B1} + R_{B2})$**, typically **0.5–0.8**.
- Emitter peak (firing) voltage: **$V_P = \eta V_{BB} + V_D$** ($V_D \approx 0.7$ V).
- Once $V_E \geq V_P$, the E–B_1 region floods with carriers → R_{B1} collapses → **negative-resistance region** between peak and valley points — the key to oscillation.

UJT relaxation oscillator: R_T charges C_T toward V_BB; at V_P the UJT fires and rapidly discharges C_T through B_1 (sharp pulse across R_1); C_T recharges and the cycle repeats.

$T \approx R_T C_T \ln[1/(1-\eta)]$, $f = 1/T$

(For $\eta = 0.63$, $T \approx R_T C_T$.) Output: sawtooth at the emitter, sharp spikes at B_1 — ideal, well-synchronised **trigger pulses for SCR/TRIAC firing**.

2.11 Triode and Vacuum-Tube Basics (brief — syllabus item)

Before semiconductors, thermionic vacuum tubes did every electronic job.

- **Diode valve (Fleming, 1904):** heated cathode (thermionic emission — Richardson's law) + anode (plate). Conducts only when plate is positive → rectifier. Space-charge-limited current follows **Child–Langmuir 3/2-power law: $I_P \propto V_P^{(3/2)}$**.
- **Triode (De Forest, 1906):** adds a **control grid** between cathode and plate. A small negative grid voltage strongly throttles plate current → **amplification**.

Triode constants (defined at constant third variable):

Constant	Definition	Typical
Amplification factor **μ**	$-\partial V_P/\partial V_G$ at constant I_P	10–100
Plate (dynamic) resistance **r_p**	$\partial V_P/\partial I_P$ at constant V_G	5–50 kΩ
Transconductance (mutual conductance) **g_m**	$\partial I_P/\partial V_G$ at constant V_P	1–10 mA/V

Master relation: $\mu = g_m \times r_p$

Triode amplifier gain: **$A = \mu R_L/(r_p + R_L)$** (approaches μ as $R_L \to \infty$).

Later tubes: **tetrode** (screen grid — but has negative-resistance kink from secondary emission), **pentode** (adds suppressor grid — removes the kink; higher gain). Tubes survive today in high-power RF transmitters, magnetrons (microwave ovens), CRTs and boutique audio. Triode ↔ semiconductor analogy: cathode↔emitter/source, grid↔base/gate, plate↔collector/drain.

2.12 Inverters and UPS

2.12.1 Inverter

An **inverter converts DC into AC** of desired voltage and frequency (the inverse of a rectifier). Power devices used: MOSFET/IGBT (modern), thyristors with forced commutation (legacy).

Classification:

Basis	Types
Source	**VSI** (Voltage Source Inverter — stiff DC voltage, capacitor at input) / **CSI** (Current Source Inverter — stiff DC current, large series inductor)
Phases	Single-phase (half-bridge: 2 switches, output ±V_dc/2; full-bridge/H-bridge: 4 switches, output ±V_dc) / Three-phase (6 switches; 120°- or 180°-conduction modes)
Output waveform	**Square wave** (simple, high harmonic content), **modified/quasi sine** (stepped), **pure sine** (PWM + LC filter — best for motors and sensitive loads)
Control	PWM — **sinusoidal PWM (SPWM)** compares a sine reference with a high-frequency triangular carrier; modulation index m_a = V_ref/V_carrier controls the fundamental output; harmonics pushed to high frequency, easily filtered

Square-wave full-bridge output: RMS = V_dc; fundamental RMS = (4V_dc/π)/√2 = **0.9 V_dc**; only odd harmonics present (3rd = 1/3 amplitude, 5th = 1/5 …). Harmonic quality measured by **THD** (square wave THD ≈ 48.3 %).

Applications: solar/battery power systems, UPS, variable-frequency drives (VFD), induction heating, HVDC receiving stations, electric traction auxiliaries.

2.12.2 UPS — Uninterruptible Power Supply

A UPS delivers continuous, conditioned AC power to critical loads (computers, servers, medical, railway signalling) during mains failure or disturbances.

Core blocks: Rectifier/charger (AC→DC), **Battery bank** (energy storage), **Inverter** (DC→AC), **Static transfer switch/bypass**.

Three standard topologies:

Type	Normal-path operation	Transfer time	Protection quality	Typical use
Offline (standby)	Load fed **directly from mains**; inverter idle, battery on float charge. On failure, static switch transfers load to inverter	~4–10 ms (brief break)	Basic — voltage sags/noise pass through in normal mode	PCs, home use; cheapest, most efficient in standby
Line-interactive	Mains feeds load through an **AVR (buck–boost autotransformer)**; the bidirectional inverter/charger sits across the line and corrects sags/swells without draining the battery	~2–4 ms	Good — regulates voltage without battery use	Small servers, network gear
Online (double-conversion)	Load is fed **always from the inverter**: mains → rectifier → DC bus (battery floats here) → inverter → load. Mains failure causes **zero transfer time** — battery is already in the path	**0 ms (no break)**	Best — total isolation from all mains disturbances; output frequency/voltage fully regenerated	Data centres, hospitals, railway signalling & telecom

Block-diagram memory aid (online UPS): Mains → [Rectifier/Charger] → DC bus ← [Battery] → [Inverter] → [Static bypass switch] → Load. The bypass carries the load if the inverter itself fails or during overload/maintenance.

2.12.3 Battery sizing basics

- **Battery Ah capacity required ≈ (Load power in W × backup time in h) / (Battery voltage × inverter efficiency × depth-of-discharge)**
- Example: 600 W load, 1 h backup, 24 V bank, η_inv = 0.85, DoD = 0.8:

Ah = 600 × 1 / (24 × 0.85 × 0.8) ≈ **36.8 Ah → choose 40–42 Ah bank (e.g. two 12 V, 42 Ah in series)**.

- **VA vs W:** UPS rating in VA; usable watts = VA × power factor (typically 0.6–0.8 for small UPS; unity for premium online units).
- Battery types: flooded/tubular lead-acid (long life, needs topping-up), **SMF/VRLA** (sealed maintenance-free — standard in UPS), Li-ion (modern, light, costly). Float charge ≈ 2.25 V/cell, boost ≈ 2.4 V/cell (lead-acid, 2 V nominal per cell → 6 cells = 12 V battery).
- Backup time ∝ Ah / load current; runtime doubles (roughly) when load halves.

2.13 Worked Numerical Examples (12)

Example 1 — Diode dynamic resistance.

A Si diode carries a forward DC current of 2 mA at 300 K. Find its dynamic resistance (η = 1).

Solution: r_d = ηV_T/I = 26 mV / 2 mA = **13 Ω**.

Example 2 — Zener regulator.

V_in = 20 V, R_S = 1 kΩ, V_Z = 10 V, R_L = 5 kΩ. Find I_S, I_L, I_Z and power dissipated in the Zener.

Solution:

I_S = (20 − 10)/1k = 10 mA; I_L = 10/5k = 2 mA; I_Z = 10 − 2 = **8 mA**.

P_Z = V_Z × I_Z = 10 × 8 mA = **80 mW**.

Example 3 — Half-wave rectifier with transformer.

A HWR is fed from 230 V, 50 Hz mains through a 10:1 step-down transformer. Diode ideal, R_L = 100 Ω. Find V_dc, I_dc, PIV and ripple frequency.

Solution:

Secondary RMS = 230/10 = 23 V → V_m = 23√2 ≈ 32.5 V.

V_dc = V_m/π = 32.5/3.1416 ≈ **10.35 V**; I_dc = 10.35/100 ≈ **103.5 mA**.

PIV = V_m ≈ **32.5 V**; ripple frequency = **50 Hz**.

Example 4 — Full-wave centre-tap numbers.

A centre-tapped FWR has 30-0-30 V (RMS) secondary, ideal diodes, R_L = 1 kΩ. Find V_dc, I_dc, PIV, and ripple factor.

Solution:

V_m (per half) = $30\sqrt{2} \approx 42.43$ V.

$V_{dc} = 2V_m/\pi = 2 \times 42.43/\pi \approx$ **27 V**; I_{dc} = 27 mA.

PIV = $2V_m \approx$ **84.85 V**; $\gamma =$ **0.482**.

Example 5 — Rectifier efficiency with diode resistance.

A half-wave rectifier uses a diode of forward resistance $r_f = 20\ \Omega$ with $R_L = 480\ \Omega$. Find the rectification efficiency.

Solution:

$\eta = 0.406/(1 + r_f/R_L) = 0.406/(1 + 20/480) = 0.406/1.04167 \approx$ **0.39 → 39 %**.

Example 6 — Capacitor filter ripple.

A bridge rectifier (50 Hz mains) delivers 20 mA DC to a load with a 1000 µF filter capacitor. Find peak-to-peak ripple voltage.

Solution:

Full-wave: $V_{r(pp)} = I_{dc}/(2fC) = 0.02/(2 \times 50 \times 1000\times10^{-6}) = 0.02/0.1 =$ **0.2 V**.

Example 7 — α to β conversion.

A transistor has $\alpha = 0.98$ and $I_E = 5$ mA. Find β, I_C and I_B.

Solution:

$\beta = \alpha/(1-\alpha) = 0.98/0.02 =$ **49**.

$I_C = \alpha I_E = 0.98 \times 5 =$ **4.9 mA**; $I_B = I_E - I_C =$ **0.1 mA** (check: $I_C/I_B = 49$ ✓).

Example 8 — Voltage-divider bias Q-point.

V_{CC} = 12 V, R_1 = 40 kΩ, R_2 = 8 kΩ, R_C = 2 kΩ, R_E = 1 kΩ, Si transistor (V_{BE} = 0.7 V), β large. Find I_C and V_{CE}.

Solution:

$V_B = 12 \times 8/(40+8) = 2$ V; $V_E = 2 - 0.7 = 1.3$ V.

$I_C \approx I_E = 1.3/1k =$ **1.3 mA**.

$V_{CE} = 12 - 1.3\text{ mA} \times (2k + 1k) = 12 - 3.9 =$ **8.1 V**. Q-point = (8.1 V, 1.3 mA).

Example 9 — Load line end-points.

For V_{CC} = 10 V, R_C = 2 kΩ (no R_E), state the DC load-line intercepts and the mid-point Q for maximum swing.

Solution:

Cut-off point: (V_{CE}, I_C) = (**10 V, 0**). Saturation point: $I_C = 10/2k =$ **5 mA** at $V_{CE} = 0$.

Mid-point Q: $V_{CEQ} = 5$ V, $I_{CQ} =$ **2.5 mA**.

Example 10 — CE amplifier gain.

A CE amplifier has I_E = 1.3 mA, R_C = 3 kΩ, R_L = 6 kΩ (capacitively coupled), emitter fully bypassed. Estimate the voltage gain.

Solution:

$r_e' = 26/1.3 = 20\ \Omega$. $R_{ac} = R_C \parallel R_L = 3k \times 6k/9k = 2\ k\Omega$.

$A_v = -R_{ac}/r_e' = -2000/20 =$ **−100** (gain 100 with 180° inversion).

Example 11 — Colpitts frequency.

A Colpitts oscillator uses C_1 = 100 pF, C_2 = 400 pF and L = 50 µH. Find the oscillation frequency.

Solution:

$C_{eq} = C_1C_2/(C_1+C_2) = 100 \times 400/500 = 80$ pF.

$f = 1/(2\pi\sqrt{(LC_{eq})}) = 1/(2\pi\sqrt{(50\times10^{-6} \times 80\times10^{-12})})$

$= 1/(2\pi\sqrt{(4\times10^{-15})}) = 1/(2\pi \times 6.325\times10^{-8}) \approx$ **2.52 MHz**.

Example 12 — UJT relaxation oscillator.

A UJT with η = 0.63 is used with R_T = 10 kΩ, C_T = 0.1 µF, V_{BB} = 12 V. Find the firing voltage and oscillation frequency (V_D = 0.7 V).

Solution:

$V_P = \eta V_{BB} + V_D = 0.63 \times 12 + 0.7 =$ **8.26 V**.

$T = R_T C_T \ln[1/(1-\eta)] = 10^4 \times 10^{-7} \times \ln(1/0.37) = 10^{-3} \times 0.994 \approx 1$ ms.

$f = 1/T \approx$ **1 kHz**.

Bonus Example 13 — Wien bridge.

R = 10 kΩ, C = 0.01 µF in a Wien bridge oscillator. $f = 1/(2\pi RC) = 1/(2\pi \times 10^4 \times 10^{-8}) = 1/(2\pi \times 10^{-4}) \approx$ **1.59 kHz**; minimum amplifier gain required = **3**.

Bonus Example 14 — Class-B efficiency check.

A class-B push-pull amplifier with V_{CC} = 20 V drives a load to a peak output of 20 V (full swing). Maximum efficiency = π/4 = **78.5 %**. If the swing is only 10 V peak, $\eta = (\pi/4)(V_{peak}/V_{CC}) = 0.785 \times 0.5 \approx$ **39.3 %**.

2.14 PYQ Corner — 22-04-2025 CBT-2 (recreated & fully solved)

PYQ 1 (TUF statement). *Which statement about Transformer Utilisation Factor is correct?*

(a) TUF of a half-wave rectifier is 0.812

(b) TUF = DC power delivered to load ÷ AC rating of transformer secondary

(c) TUF of a bridge rectifier is less than that of a half-wave rectifier

(d) TUF is always greater than 1

Solution: TUF is defined as P_dc/(transformer VA rating) — option **(b)**. Values: HWR 0.287, FW-CT 0.693, bridge 0.812 (highest); TUF < 1 always. Hence (a), (c), (d) are all wrong. **Answer: (b)**

PYQ 2 (BJT current relation). *For a junction transistor, which of the following is always true?*

(a) $I_C = I_E + I_B$ (b) $I_B = I_C + I_E$ (c) **$I_E = I_C + I_B$** (d) $I_E = I_C - I_B$

Solution: KCL at the device: the emitter current splits into collector and base components: **$I_E = I_C + I_B$**. Numerically I_E is the largest, I_B the smallest ($I_B \approx$ 1–5 % of I_E). **Answer: (c)**

PYQ 3 (HWR with transformer — numerical). *A half-wave rectifier is supplied from 220 V, 50 Hz mains through a step-down transformer of turns ratio 10:1. Assuming an ideal diode, the DC output voltage is approximately:*

(a) 22 V (b) 9.9 V (c) 31.1 V (d) 14 V

Solution: Secondary RMS = 220/10 = 22 V → $V_m = 22\sqrt{2} = 31.11$ V.

$V_{dc} = V_m/\pi = 31.11/3.1416 \approx$ **9.9 V**. **Answer: (b)**

(Trap options: 22 V is the RMS, 31.1 V is the peak, 14 V ≈ 0.636×22 is the *full-wave* value.)

PYQ 4 (Bridge vs centre-tap). *Which of the following is NOT an advantage of the bridge rectifier over the centre-tapped full-wave rectifier?*

(a) No centre-tapped transformer is required

(b) PIV rating required per diode is half

(c) Higher transformer utilisation factor

(d) Only one diode drop occurs in the conduction path

Solution: In a bridge, **two** diodes conduct in series each half-cycle (≈ 1.4 V total drop); the centre-tap circuit has only one. So (d) is false — it is actually a *disadvantage*. (a), (b), (c) are genuine advantages. **Answer: (d)**

PYQ 5 (Colpitts — incorrect statement). *Which statement about the Colpitts oscillator is INCORRECT?*

(a) It uses a capacitive voltage divider for feedback

(b) Its frequency is $f = 1/(2\pi\sqrt{(LC_{eq})})$ with $C_{eq} = C_1C_2/(C_1+C_2)$

(c) It uses a tapped inductor for feedback

(d) It is an LC (tuned) oscillator suitable for RF generation

Solution: The tapped-**inductor** feedback belongs to the **Hartley** oscillator; Colpitts uses the tapped-**capacitor** pair C_1–C_2. Mnemonic: **C**olpitts–**C**apacitor, **H**artley–**H**enry. **Answer: (c)**

PYQ 6 (n-type majority carriers). *In an n-type semiconductor, the majority charge carriers are:*

(a) Holes (b) **Electrons** (c) Protons (d) Negative ions

Solution: Pentavalent donor atoms (P, As, Sb) contribute one free electron each → electrons are majority carriers, holes minority. The crystal remains electrically neutral. **Answer: (b)**

PYQ 7 (Ideal diode, V(t) = $\sin^2 t + \cos^2 t$). *An ideal diode is connected in series with a 1 kΩ load across a source V(t) = $\sin^2 t + \cos^2 t$ volts. The current through the load is:*

(a) Zero (b) 1 mA DC (c) 1 mA AC at 1 rad/s (d) 0.5 mA pulsating

Solution: Trigonometric identity: $\sin^2 t + \cos^2 t$ = **1** for all t → the source is a pure **1 V DC** supply. The ideal diode (zero drop) is forward biased permanently, so I = 1 V / 1 kΩ = **1 mA steady DC**; output voltage across the load = 1 V DC. **Answer: (b)**

Lesson: RRB hides a mathematics identity inside an electronics question — always simplify the source expression first.

PYQ 8 (Oscillator — false statement). *Which statement about oscillators is FALSE?*

(a) The Barkhausen criterion requires |Aβ| = 1 and total loop phase shift of 0° or 360°

(b) An oscillator needs an external AC input signal to sustain its output

(c) Oscillations start from internal noise or the switching transient

(d) If |Aβ| < 1, the oscillations die out

Solution: An oscillator is precisely a circuit that generates AC **without** any external input — positive feedback recycles a fraction of the output as its own input, with internal noise providing the seed. Statement (b) is false; (a), (c), (d) are correct. **Answer: (b)**

2.15 Practice MCQs (30)

Q1. The forbidden energy gap of silicon at room temperature is about:

(a) 0.3 eV (b) 0.72 eV (c) 1.1 eV (d) 6 eV

Q2. The barrier potential of a germanium PN junction at 25 °C is approximately:

(a) 0.7 V (b) 0.3 V (c) 1.1 V (d) 0.1 V

Q3. With every 10 °C rise in temperature, the reverse saturation current of a diode:

(a) halves (b) doubles (c) stays constant (d) becomes zero

Q4. In a p-type semiconductor, the Fermi level lies:

(a) closer to the conduction band (b) closer to the valence band (c) exactly at mid-gap (d) inside the conduction band

Q5. Zener breakdown (as distinct from avalanche) occurs in diodes that are:

(a) lightly doped (b) heavily doped with thin depletion layers (c) undoped (d) metal–semiconductor junctions

Q6. A varactor diode is operated in:

(a) forward bias (b) reverse bias (c) breakdown (d) zero bias only

Q7. The negative-resistance region in a tunnel diode lies between:

(a) zero and peak point (b) peak and valley points (c) valley point and infinity (d) it has no such region

Q8. The ripple factor of an ideal full-wave rectifier is:

(a) 1.21 (b) 0.482 (c) 0.812 (d) 1.11

Q9. For a half-wave rectifier with peak input V_m, the PIV rating of the diode must be at least:

(a) $V_m/2$ (b) V_m (c) $2V_m$ (d) πV_m

Q10. The lowest ripple frequency at the output of a bridge rectifier on 50 Hz mains is:

(a) 25 Hz (b) 50 Hz (c) 100 Hz (d) 150 Hz

Q11. A shunt-capacitor filter gives the best performance when:

(a) load current is large (b) load current is small (c) frequency is very low (d) capacitor is very small

Q12. A clamper circuit changes the:

(a) peak-to-peak value (b) waveshape (c) DC level of the waveform (d) frequency

Q13. In a BJT, the emitter region compared to base and collector is:

(a) most lightly doped (b) most heavily doped (c) largest in area (d) thinnest

Q14. If $\alpha = 0.99$, β equals:

(a) 9 (b) 49 (c) 99 (d) 100

Q15. The configuration with voltage gain less than unity but highest input impedance is:

(a) CB (b) CE (c) CC (d) all equal

Q16. For a transistor used as an amplifier, the junction bias condition is:

(a) both junctions forward (b) both reverse (c) EB forward, CB reverse (d) EB reverse, CB forward

Q17. The stability factor of a fixed-bias circuit is:

(a) 1 (b) $1 + \beta$ (c) $1/(1+\beta)$ (d) β

Q18. In a voltage-divider-biased amplifier, the emitter bypass capacitor is used to:

(a) increase DC stability (b) prevent AC degeneration and restore gain (c) block collector DC (d) reduce input impedance

Q19. The device with the highest input impedance among these is:

(a) BJT-CE (b) JFET (c) MOSFET (d) BJT-CB

Q20. Shockley's equation $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ applies to:

(a) BJT (b) JFET (c) SCR (d) UJT

Q21. Maximum theoretical efficiency of a class-B push-pull amplifier is:

(a) 25 % (b) 50 % (c) 78.5 % (d) 100 %

Q22. Crossover distortion is associated with class:

(a) A (b) B (c) C (d) AB

Q23. Negative feedback in an amplifier:

(a) increases gain and decreases bandwidth (b) decreases gain and increases bandwidth (c) increases both (d) decreases both

Q24. An RC phase-shift oscillator using three identical RC sections needs each section to provide a phase shift of:

(a) 30° (b) 45° (c) 60° (d) 90°

Q25. The minimum amplifier gain required for a Wien bridge oscillator is:

(a) 1 (b) 3 (c) 29 (d) 10

Q26. The oscillator with the best frequency stability is:

(a) Hartley (b) Colpitts (c) RC phase-shift (d) crystal

Q27. A monostable multivibrator has:

(a) no stable state (b) one stable state (c) two stable states (d) three stable states

Q28. An SCR is a:

(a) 2-layer, 1-junction device (b) 3-layer, 2-junction device (c) 4-layer, 3-junction device (d) 5-layer, 4-junction device

Q29. The relation between latching current I_L and holding current I_H of an SCR is:

(a) $I_L = I_H$ (b) $I_L < I_H$ (c) $I_L > I_H$ (d) unrelated

Q30. In an online (double-conversion) UPS, the transfer time on mains failure is:

(a) 10 ms (b) 4 ms (c) 25 ms (d) zero

Answer Key with one-line explanations

Q	Ans	One-line reason
1	c	Si Eg ≈ 1.1 eV (Ge 0.72 eV, diamond ≈ 6 eV)
2	b	Ge barrier ≈ 0.3 V; Si ≈ 0.7 V
3	b	I_0 doubles per 10 °C rise (minority-carrier generation)
4	b	Acceptor level/holes pull the Fermi level toward the valence band
5	b	Heavy doping → thin depletion → field tunnelling (Zener), V_Z < ~5–6 V
6	b	Reverse bias varies depletion width → voltage-variable capacitance
7	b	Current falls from peak (~65 mV) to valley → negative resistance
8	b	$\gamma(FW) = \sqrt{(1.11^2 - 1)} = 0.482$; 1.21 is half-wave
9	b	HWR diode blocks only the peak V_m; 2V_m applies to centre-tap FWR
10	c	Full-wave ripple frequency = 2f = 100 Hz
11	b	$\gamma = 1/(4\sqrt{3}\, fCR_L)$: small I_dc (large R_L) → low ripple
12	c	Clamper shifts the DC level; shape and p-p value unchanged
13	b	Emitter is doped heaviest to inject carriers efficiently
14	c	$\beta = \alpha/(1-\alpha) = 0.99/0.01 = 99$
15	c	CC (emitter follower): A_v < 1, very high Z_in — buffer
16	c	Active region: EB forward, CB reverse
17	b	Fixed bias: S = 1 + β — the worst stability
18	b	Bypassing R_E removes AC negative feedback → full gain, DC stability kept
19	c	MOSFET's insulated gate: 10^{10}–10^{15} Ω
20	b	JFET square-law transfer characteristic
21	c	η_{max}(class B) = π/4 = 78.5 %
22	b	Class B dead zone near ±0.7 V; class AB cures it
23	b	$A_f = A/(1+A\beta)$; BW multiplies by (1+Aβ) — G×BW constant
24	c	3 × 60° = 180° from network + 180° from CE amplifier = 360°
25	b	Lead–lag network attenuates by 3 → gain ≥ 3

Q	Ans	One-line reason
26	d	Quartz Q ~10^4–10^6 → ppm-level stability
27	b	Monostable = one-shot: one stable + one quasi-stable state
28	c	SCR = PNPN: 4 layers, 3 junctions, 3 terminals
29	c	I_L (to latch ON) ≈ 2–3 × I_H (to stay ON)
30	d	Inverter is always in the load path → zero-break transfer

2.16 Rapid-Revision Formula Sheet

Semiconductors & diode

- Mass action: $n \cdot p = n_i^2$ $\sigma = e(n\mu_n + p\mu_p)$ Einstein: $D/\mu = V_T = kT/e \approx$ **26 mV @ 300 K**
- Diode: $I = I_0(e^{(V/\eta V_T)} - 1)$; $r_d = \eta V_T/I \approx 26$ mV/I(mA)
- Barrier: Si 0.7 V, Ge 0.3 V; falls ≈ 2 mV/°C; I_0 doubles per 10 °C
- Zener breakdown: heavy doping, $V_Z < 5$–6 V, −ve temp-coeff; Avalanche: light doping, $V_Z > 6$ V, +ve temp-coeff
- LED wavelength: λ(nm) = 1240/Eg(eV)

Rectifiers (V_m = peak input)

	HW	FW-CT	Bridge
V_{dc}	V_m/π (0.318V_m)	$2V_m/\pi$ (0.636V_m)	$2V_m/\pi$
I_{rms}	$I_m/2$	$I_m/\sqrt{2}$	$I_m/\sqrt{2}$
FF	1.57	1.11	1.11
γ	1.21	0.482	0.482
η_{max}	40.6 %	81.2 %	81.2 %
PIV	V_m	$2V_m$	V_m
TUF	0.287	0.693	0.812
f_{ripple}	f	2f	2f

- η(HW) = 0.406/(1 + r_f/R_L); η(FW) = 0.812/(1 + r_f/R_L)
- γ = $\sqrt{(FF^2 - 1)}$; C-filter: γ = $1/(4\sqrt{3}\, fCR_L)$ (FW); $V_{r(pp)} = I_{dc}/(2fC)$ (FW)

BJT

- **$I_E = I_C + I_B$**; α = I_C/I_E; β = I_C/I_B; γ = 1 + β
- **β = α/(1−α); α = β/(1+β)**; $I_{CEO} = (1+\beta)I_{CBO}$
- Load line: $V_{CE} = V_{CC} - I_C(R_C + R_E)$; $I_C(sat) = V_{CC}/(R_C+R_E)$
- $r_e' = 26$ mV/I_E(mA); CE gain $A_v = -(R_C \parallel R_L)/r_e'$
- Stability: fixed bias S = 1+β (worst); voltage-divider S ≈ 1 + R_B/R_E (best)

FET

- $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$; $g_m = g_{m0}(1 - V_{GS}/V_P)$; $g_{m0} = 2I_{DSS}/|V_P|$; $\mu = g_m r_d$

Amplifier classes

- A: 25 % (50 % transformer) — 360°; B: **78.5 %** — 180°; AB: between; C: > 90 % — < 180°
- Feedback: A_f = A/(1+Aβ); BW_f = (1+Aβ)BW; distortion ÷ (1+Aβ)

Oscillators

- Barkhausen: |Aβ| = 1, phase = 0°/360°
- RC phase-shift: f = 1/(2πRC√6), gain ≥ 29
- Wien: f = 1/(2πRC), gain ≥ 3
- Hartley: f = 1/(2π√((L₁+L₂+2M)C)) — tapped inductor
- Colpitts: f = 1/(2π√(L·C₁C₂/(C₁+C₂))) — tapped capacitor
- Clapp: f ≈ 1/(2π√(LC₃)), C₃ smallest
- Crystal: f_s = 1/(2π√(LC)); inductive between f_s and f_p; Q ~ 10^4–10^6

Multivibrators / UJT / SCR

- Astable (BJT): T = 0.693(R_1C_1 + R_2C_2); symmetric T = 1.386RC
- Monostable: T = 0.693RC; n flip-flops → MOD 2^n counter (÷2 per FF)
- UJT: V_P = ηV_BB + 0.7; T = RC ln[1/(1−η)]; η = R_B1/(R_B1+R_B2) ≈ 0.5–0.8
- SCR: I_A = I_leak/[1 − (α₁+α₂)]; latches when α₁+α₂ → 1; I_L > I_H

Triode

- **μ = g_m × r_p**; gain A = μR_L/(r_p + R_L); Child's law I_P ∝ V_P^(3/2)

Inverter / UPS

- Square-wave fundamental (full-bridge): V_1(rms) = 0.9 V_dc; odd harmonics only
- UPS transfer times: offline 4–10 ms > line-interactive 2–4 ms > **online 0 ms**
- Battery Ah = (W × hours)/(V_batt × η_inv × DoD); Watts = VA × pf

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End of Chapter 2. Next: Chapter 3 — Digital Electronics.